

Multimodal Deep Learning for Appendicitis Diagnosis: Integrating Clinical and Ultrasound Data

Saroj Baral, *sb2ek@mtmail.mtsu.edu*

Abstract—Accurate diagnosis of acute appendicitis remains challenging. To improve diagnostic performance, I developed a multimodal deep learning model combining clinical tabular data and ultrasound images. Tabular features were represented using a multilayer perceptron, while a fine-tuned ResNet-50 represented ultrasound images. An intermediate fusion strategy integrated both modalities into a joint representation for classification. Using 10-fold cross-validation, the multimodal model achieved a mean Area Under the Receiver Operating Characteristic Curve (AUROC) of 0.86, outperforming a baseline tabular-only model (AUROC 0.79). These findings highlight the promise of multimodal deep learning in enhancing diagnostic accuracy.

Index Terms—multimodal, appendicitis

I. INTRODUCTION

THE inflammation of the appendix, known as acute appendicitis, constitutes a primary reason for emergency abdominal surgery worldwide, predominantly impacting younger individuals [1], [2]. Despite its common occurrence, achieving a prompt and precise diagnosis remains a significant clinical challenge. The gravity of appendicitis arises not merely from its prevalence but from the substantial risks associated with diagnostic delays or errors. Chief among these risks is appendix perforation, which can escalate to severe complications like peritonitis, abscess formation, extended hospital stays, heightened morbidity, and potentially, mortality [1].

Conventionally, the diagnostic pathway integrates patient medical history, clinical examination findings, laboratory test results, and frequently, clinical scoring tools [3], [4]. Nevertheless, the archetypal symptom progression—migrating pain to the right lower abdomen, loss of appetite, nausea, and fever—is notably absent in as many as 40% of cases, rendering clinical judgment alone inadequate for reliable diagnosis [5]. This diagnostic ambiguity forces a challenging clinical trade-off: postponing diagnosis elevates the likelihood of perforation, whereas proceeding with surgery based purely

on suspicion contributes to a considerable rate of negative appendectomies (the surgical removal of a healthy appendix). Such unnecessary surgeries entail inherent risks and inflate healthcare costs. Furthermore, while scoring systems assist in risk assessment, they frequently fall short of the sensitivity and specificity needed for conclusive identification across varied patient groups [6], [7].

To enhance diagnostic precision, medical imaging is often utilized. Ultrasound (US) is broadly advocated as the initial imaging method, particularly for children and pregnant women, owing to its non-invasive character, absence of ionizing radiation, and comparatively lower expense than computed tomography (CT) [8], [9]. Yet, the diagnostic efficacy of US heavily relies on the operator's expertise, and its interpretation can be subjective, resulting in notable variability between observers, even seasoned radiologists. This issue is especially pronounced when the appendix is not visualized or presents ambiguously [10]. Factors such as the patient's body size and obscuring intestinal gas can further impede ultrasound's utility. Although CT provides superior sensitivity and specificity, its application involves a significant radiation dose, posing a specific concern for younger patients [9].

The inherent shortcomings in clinical evaluation, scoring methodologies, and imaging interpretation underscore the necessity for more objective, dependable, and accurate diagnostic instruments. Artificial intelligence (AI) and machine learning (ML) represent promising technological advancements capable of addressing these diagnostic hurdles in medicine [11]. By discerning intricate patterns within extensive datasets, ML models possess the capacity to detect subtle disease indicators that might elude consistent human observation. The creation of computer-aided diagnosis (CADx) systems, driven by AI, seeks to furnish clinicians with quantitative, evidence-based insights, thereby supporting and refining their diagnostic processes, potentially diminishing variability and boosting accuracy [12].

Within the appendicitis diagnostic sphere, AI research has typically focused on analyzing either clinical data or imaging data in isolation. However, effective clinical diagnosis frequently necessitates the synthesis of information from diverse sources; a patient's history, physical examination, lab results, and imaging scans are usually evaluated collectively. This reality motivates the exploration of a *multimodal* machine learning paradigm [13]. Multimodal models, specifically engineered to process and integrate data from disparate types (like structured tabular information and unstructured image data), offer the potential to construct a more comprehensive clinical picture. By harnessing the unique advantages of varied data inputs,

Manuscript received July 10, 2012. This paper is an expanded paper from the IEEE MTT-S Int. Microwave Symposium held on June 17-22, 2012 in Montreal, Canada. This work was funded in part by the Office of Naval Research under the Defense Advanced Research Projects Agency (DARPA) Microscale Power Conversion (MPC) Program under Grant N00014-11-1-0931, and in part by the Advanced Research Projects Agency-Energy (ARPA-E), U.S. Department of Energy, under Award Number DE-AR0000216.

M. Roberg is with TriQuint Semiconductor, 500 West Renner Road Richardson, TX 75080 USA (e-mail: michael.roberg@tqs.com).

T. Reveyrand is with the XLIM Laboratory, UMR 7252, University of Limoges, 87060 Limoges, France (e-mail: tibault.reveyrand@xlim.fr).

I. Ramos and Z. Popovic are with the Department of Electrical, Computer and Energy Engineering, University of Colorado, Boulder, CO, 80309-0425 USA (e-mail: ignacio.ramos@colorado.edu; zoya.popovic@colorado.edu).

E. Falkenstein is with Qualcomm Inc., 6150 Lookout Road Boulder, CO 80301 USA (e-mail: erez.falkenstein@gmail.com).

these models might reveal complex correlations and attain diagnostic capabilities superior to those reliant on a single data modality [14].

Consequently, the objective of this research is to develop and assess a novel multimodal deep learning architecture expressly designed for appendicitis diagnosis. The core strategy involves synergistically combining readily accessible tabular clinical data (encompassing patient demographics, symptoms, laboratory findings) with corresponding abdominal ultrasound images. By employing specialized neural network structures for each data type and subsequently merging their learned feature representations, the integrated model is expected to achieve enhanced diagnostic accuracy relative to models trained solely on one modality. This methodology aims to produce a more reliable diagnostic aid, rigorously evaluated using AUROC on a publicly available dataset, with the goal of improving patient outcomes through more precise and timely appendicitis diagnosis.

II. BACKGROUND

The diagnostic pathway for suspected appendicitis has evolved considerably. Initial efforts focused on standardizing clinical assessment through scoring systems like the Alvarado score [3], [6] and the Pediatric Appendicitis Score [4], among others [15]. While valuable for risk stratification, these scores often lack definitive discriminatory power [7], [15]. This limitation prompted the exploration of machine learning (ML) techniques leveraging electronic health record data. Early ML studies demonstrated the potential of models such as random forests, SVMs, and neural networks using only tabular clinical and laboratory data [16], [17]. These studies, often employing models well-suited for structured data, also highlighted ongoing questions regarding the optimal approach for tabular inputs—particularly the comparison between deep learning methods, such as Multilayer Perceptrons (MLPs), and traditional machine learning techniques, like gradient boosted trees, given the challenges deep learning can face with heterogeneous tabular features [18], [19].

Parallel to advancements in analyzing clinical data, the diagnostic value of imaging, particularly ultrasound, gained recognition. Consequently, subsequent research focused on incorporating imaging findings. Notably, Marcinkevičs et al. demonstrated improved diagnostic performance by adding *structured* ultrasound findings—features manually coded from expert interpretations—to clinical data models [20]. However, this reliance on manual feature extraction presents a barrier to full automation. A key goal in current research, including this study, is therefore to leverage raw medical images directly using deep learning, bypassing the need for explicit expert feature annotation.

The advent of deep learning, particularly Convolutional Neural Networks (CNNs), has significantly transformed medical image analysis [21], [22]. Architectures such as the Residual Network—a variant of CNN that uses residual connections—have proven highly effective, with ResNet-50, a 50-layer convolutional neural network [23], being a widely adopted example. Using a ResNet-50 model pre-trained on

large benchmark datasets (e.g., ImageNet) enables the transfer of learned representations, which is especially valuable in medical imaging tasks where annotated datasets are often limited. The model can then be fine-tuned for specific medical applications to better adapt its features [22]. In the context of appendicitis diagnosis, later work by Marcinkevičs et al. employed modified Concept Bottleneck Models [24], built on CNNs, achieving strong performance using ultrasound images alone while also promoting interpretability [25].

These parallel advancements in analyzing both clinical tabular data and medical images naturally motivate the field of *multimodal machine learning*, which seeks to integrate information from heterogeneous data sources [13], [11]. Developing effective multimodal models, however, introduces distinct challenges, primarily concerning how data from different modalities are represented and subsequently fused [26].

A primary challenge in multimodal ML lies in representation learning: how to effectively process and create meaningful feature representations from diverse data sources. Different modalities (e.g., tabular clinical records, ultrasound images) possess fundamentally different structures, dimensionalities, and statistical properties. Broadly, representation learning approaches are categorized into two types: *Joint Representation* and *Coordinated Representation* [13]. In Joint Representation, different modalities are represented in a single feature space, while in Coordinated Representation, each modality exists in a separate space, coordinating with one another (often using cosine similarity).

Another key challenge in multimodal learning is *information fusion*, which involves effectively combining representations learned from different data modalities [14]. One important consideration is the stage at which fusion occurs—whether at the input level (early fusion), output level (late fusion), or within the network (intermediate fusion). Among these, *intermediate fusion* is the most commonly adopted approach [27], [13], as it merges feature representations generated by modality-specific encoders, thus balancing specialized feature extraction with the ability to learn complex cross-modal interactions before final prediction. This study adopts the intermediate fusion approach.

Within intermediate fusion, several mechanisms can be employed to combine latent feature vectors. A simple yet effective baseline is *concatenation*, which appends vectors and preserves all available information, leaving it to subsequent layers to model inter-modal interactions [28]. More adaptive methods include *gated* or *weighted fusion* strategies, such as using learned scalar weights or attention-based mechanisms to control the relative influence of each modality during fusion [14], [13]. Particularly in biomedical domains, joint representations derived through intermediate fusion have been shown to outperform coordinated representations [26].

Building on these foundations, this study develops a multimodal deep learning model for appendicitis diagnosis. It addresses the representation challenge by employing a MLP for tabular data and a fine-tuned pre-trained ResNet-50 for ultrasound images, each producing modality-specific feature vectors. It then addresses the fusion challenge by exploring intermediate fusion strategies, including both concatenation

and weighted sum methods, to combine the learned representations.

III. METHODS

A. Data Preprocessing

The dataset sourced from [25], comprises 579 pediatric patients admitted to Children's Hospital St. Hedwig, each associated with 1 to 15 abdominal ultrasound images capturing regions such as the right lower quadrant, appendix, intestines, and lymph nodes. The dataset is balanced, with 251 patients diagnosed with appendicitis and the remainder classified as non-appendicitis. Two data modalities—tabular and image-based—are preprocessed independently to prepare them for model input. The dataset is publicly available at <https://zenodo.org/records/7711412>.

1) *Tabular Data Preprocessing*: The dataset comprises tabular information, including patient demographics (age, sex, height, weight), established scoring metrics such as the Alvarado score [6] and Pediatric Appendicitis Score [4], various clinical symptoms like migratory pain and bowel characteristics, and laboratory results derived from hemograms. Notably, variables requiring expert interpretation of ultrasound images were deliberately omitted to focus predictions on readily available data.

An initial analysis identified missing values, prompting the removal of features where over 30% of the data was absent. To prevent potential data leakage during the imputation process, the dataset was split into training and testing subsets beforehand. Missing numerical entries were subsequently addressed using an iterative imputation method; this technique estimates missing values based on other features within the dataset, using the median value as an initial estimate. For missing categorical data, a K-Nearest Neighbors (KNN) imputer [29] was utilized, assigning categories by identifying the most frequent category among the nearest neighboring data points.

2) *Image Data Preprocessing*: The accompanying ultrasound images contained scanner-generated annotations peripheral to the primary region of interest. These annotations were removed by cropping. A binary mask was then created for each image using Otsu's adaptive thresholding technique [30] to effectively isolate the relevant diagnostic area. After applying this mask universally, all processed images were standardized to a 400×400 pixel resolution to ensure consistent input dimensions for the analytical model.

B. Model Architecture

The multimodal neural network is illustrated in Figure 1 taking inspiration from successful approaches in medical imaging challenges, such as Ha et al.'s work on Melanoma classification [31]. Key difference is using ResNet-50 as the image backbone and incorporating a permutation-invariant aggregation strategy to handle the variable number of ultrasound pictures per patient.

The architecture consists of four main components. First, a **tabular data encoder** processes patient metadata, including clinical and laboratory features, using a MLP. At the same time, an **image encoder** extracts features from ultrasound

images using a fine-tuned ResNet-50 model and aggregates features across multiple images for each patient. Next, a **feature fuser** integrates the intermediate representations obtained from both the tabular and image encoders. Finally, a **classifier** leverages the fused features to predict the diagnostic outcome.

1) *Tabular Data Encoder*: The tabular data encoder is a MLP that processes the 33 input clinical and laboratory features (x_{tab}). It includes two hidden layers with H_1 and H_2 neurons, respectively. The outputs of these layers are denoted by h_1 and h_2 , and are computed using ReLU activations. A final linear layer produces a tabular feature vector E_{tab} of dimension H_{tab} , to which layer normalization is applied. Mathematically, the process can be summarized as:

$$h_1 = \text{ReLU}(\text{Linear}_{H_1}(x_{\text{tab}})) \quad (1)$$

$$h_2 = \text{ReLU}(\text{Linear}_{H_2}(h_1)) \quad (2)$$

$$E_{\text{tab}} = \text{LayerNorm}(\text{ReLU}(\text{Linear}_{H_{\text{tab}}}(h_2))) \in R^{H_{\text{tab}}} \quad (3)$$

2) *Image Encoder*: The image encoder first processes each preprocessed ultrasound image (x_{img}) using a fine-tuned ResNet-50 backbone [32], producing a 1000-dimensional feature vector:

$$f_{\text{img}} = \text{ResNet-50}(x_{\text{img}}) \in R^{1000} \quad (4)$$

For patients with multiple images, the set of resulting image-level vectors $\{f_{\text{img},i}\}$ is aggregated into a single patient-level vector using element-wise summation. This operation ensures permutation invariance (output independence from image order), following principles like those in Deep Sets [33]:

$$F_{\text{summed}} = \sum_i f_{\text{img},i} \in R^{1000} \quad (5)$$

This aggregated vector (F_{summed}) is then passed through a custom MLP head (two linear layers with dimensions H_3 and H_{img} , ReLU activations and final layer normalization) to produce the final patient-level image embedding E_{img} :

$$h_{\text{head}} = \text{ReLU}(\text{Linear}_{H_3}(F_{\text{summed}})) \quad (6)$$

$$E_{\text{img}} = \text{LayerNorm}(\text{ReLU}(\text{Linear}_{H_{\text{img}}}(h_{\text{head}}))) \in R^{H_{\text{img}}} \quad (7)$$

3) *Feature Fusion*: The tabular features (E_{tab}) and the aggregated image features (E_{img}) are combined based on the fusion strategy hyperparameter S . If $S = \text{concat}$, the features are concatenated:

$$E_{\text{fused}} = \text{Concat}(E_{\text{tab}}, E_{\text{img}}) \quad (8)$$

If $S = \text{weighted_sum}$, the features are combined using learnable scalar weights ($w_{\text{tab}}, w_{\text{img}}$), normalized via softmax:

$$E_{\text{fused}} = w_{\text{tab}}E_{\text{tab}} + w_{\text{img}}E_{\text{img}} \quad (9)$$

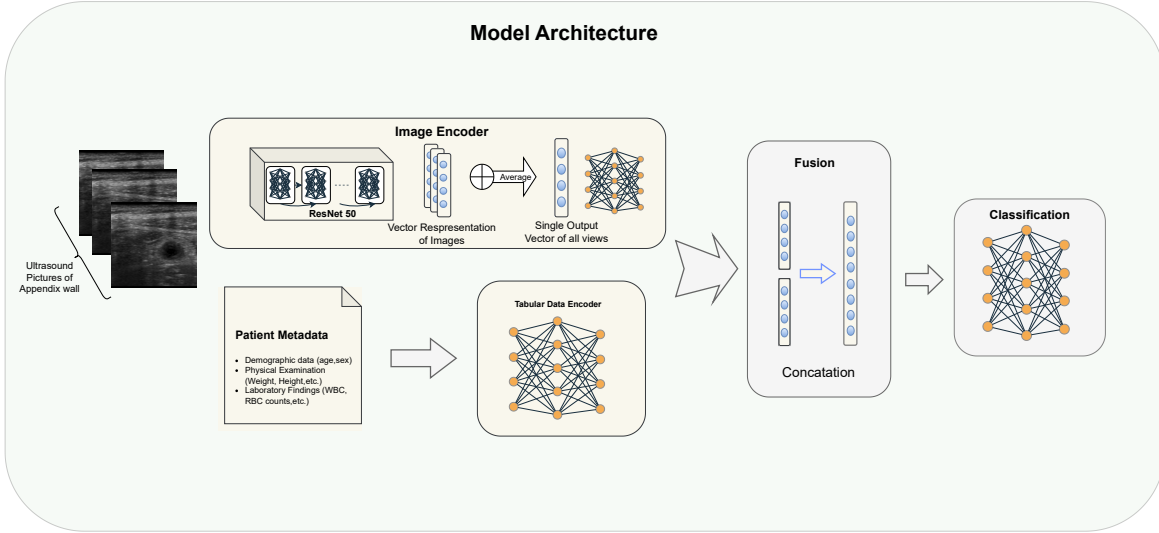


Fig. 1. **Model Architecture:** The model consists of an image encoder for ultrasound images and a tabular data encoder for patient metadata. Their outputs are fused before being passed to a final classifier.

4) *Final Classifier:* The fused feature vector (E_{fused}) is input to the final classifier, an MLP comprising two hidden layers (with H_4 and H_5 neurons and ReLU activations) and a single output neuron using a sigmoid activation function to produce the final binary classification probability (y):

$$h_1 = \text{ReLU}(\text{Linear}_{H_4}(E_{\text{fused}})) \quad (10)$$

$$h_2 = \text{ReLU}(\text{Linear}_{H_5}(h_1)) \quad (11)$$

$$y = \text{Sigmoid}(\text{Linear}_1(h_2)) \quad (12)$$

IV. EVALUATION

The model is evaluated using 10-fold cross-validation, where the dataset is partitioned into 10 subsets. In each fold, 9 subsets are used for training, and the remaining subset is used for testing. The primary metric for evaluation is the average AUROC across all folds, with the standard error reported to assess performance variability. AUROC measures the model's ability to distinguish between classes at various classification thresholds, making it a more reliable metric than accuracy. A perfect model achieves an AUROC of 1, while a model that classifies randomly yields an AUROC of 0.5.

V. EXPERIMENTAL SETUP

The multimodal model was trained for 50 epochs using binary cross-entropy loss and optimized with the Adam optimizer [34]. A 10-fold cross-validation strategy was employed for evaluation. Hyperparameters were tuned using a nested cross-validation approach on the first outer fold; specifically, this fold was further divided into 5 sub-folds for an inner cross-validation grid search to identify the optimal configuration based on performance within that fold. The specific hyperparameters explored during this grid search are detailed in Table I.

The fixed hidden layer sizes within the architecture were set based on preliminary experiments as follows: $H_1 = 256$, $H_2 = 128$ (tabular encoder); $H_3 = 500$ (image encoder

head); and $H_4 = 128$, $H_5 = 64$ (final classifier). The output dimensions of the modality encoders (H_{tab} and H_{img}) were determined by the hyperparameter search (see Table I) and were constrained to be equal ($H_{\text{tab}} = H_{\text{img}}$). A batch size (B) of 4 was used during training; this small size was chosen due to the limited overall dataset size, facilitating more frequent model updates per epoch.

For comparison purposes, a 'Tabular Only' baseline model was implemented. This model processed the patient metadata using an identical tabular data encoder structure (including hidden layers H_1, H_2 and the tuned output dimension H_{tab}) as the multimodal model. The output of this encoder was directly connected to a classification head consisting of a single output neuron with a sigmoid activation function. This baseline model was trained for the same number of epochs (50), using the same binary cross-entropy loss function and Adam optimizer, and utilized the corresponding tuned hyperparameters (e.g., learning rate, weight decay, H_{tab}) from the multimodal search where applicable. Crucially, the Tabular Only model was trained and evaluated using the exact same 10-fold cross-validation splits as the multimodal model to ensure a fair comparison. The code is available at <https://github.com/nelumbus-cloud/appendicitis>.

VI. RESULTS

A hyperparameter search identified the following optimal hyperparameters: a learning rate (η) of 0.0001, a weight decay (λ) of 0.01, concatenation ($S = \text{concat}$) as the fusion strategy, and an encoder output dimension ($H_{\text{tab}} = H_{\text{img}}$) of 50. The relatively high value selected for weight decay (0.01) suggests that significant regularization was beneficial for optimizing model performance on the validation data, potentially reflecting characteristics of the training process such as the dataset size or batch size employed.

Figure 2 presents the ROC curves obtained from 10-fold cross-validation on the test-train splits of the multimodal

Hyperparameter	Values Searched	Description / Rationale
Learning Rate (η)	{0.01, 0.001, 0.0001}	Step size for the Adam optimizer.
Weight Decay (λ)	{0.1, 0.01, 0.001}	L2 regularization parameter to mitigate overfitting.
Fusion Strategy (S)	{concat, weighted sum}	Method chosen for combining features in the intermediate fusion step.
Encoder Output Dimension ($H_{\text{tab}} = H_{\text{img}}$)	{20, 50}	Controls the dimensionality of the joint representation of both modality.

TABLE I
HYPERPARAMETERS TUNED VIA GRID SEARCH USING INNER CROSS-VALIDATION ON THE FIRST FOLD

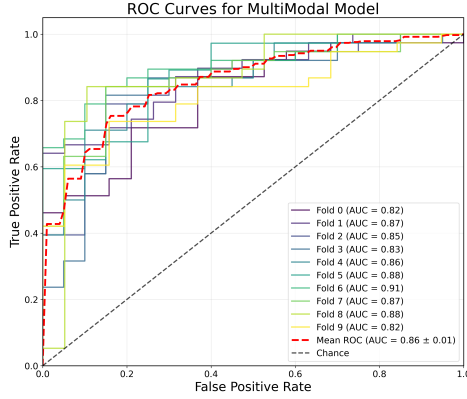


Fig. 2. **ROC Curves for MultiModal Model:** The plot displays the True Positive Rate against the False Positive Rate for the multimodal model across 10 cross-validation folds (solid colored lines). The mean ROC curve averaged across folds is shown as a dashed red line. The legend provides the Area Under the Curve (AUROC) for each fold and the mean AUROC (0.86 ± 0.01). The diagonal dashed grey line represents chance performance (AUROC = 0.5)

model, using the optimized hyperparameters. The ROC curve plots the *True Positive Rate* (TPR) against the *False Positive Rate* (FPR) at various classification thresholds. Here, the x-axis (FPR) indicates the proportion of negative cases incorrectly classified as positive, while the y-axis (TPR) reflects the proportion of actual positive cases correctly identified. A curve closer to the top-left corner indicates better classification performance, while the diagonal line represents random guessing (AUROC = 0.5). Each solid colored line corresponds to an individual fold, showing some variability in performance, with AUROC values ranging from 0.82 (Folds 0 and 9) to 0.91 (Fold 6), as indicated in the legend. The bold mean ROC curve demonstrates strong overall performance with an average AUROC of 0.86, confirming that the multimodal model performs substantially better than chance.

Figure 3 compares the average performance of the multimodal model against the tabular-only baseline using mean ROC curves derived from the 10-fold cross-validation, along with their 95% confidence intervals. The curve and confidence band for the multimodal model (blue) are positioned consistently above those of the tabular-only model (red). The clear separation between the confidence bands across the majority of the curve provides strong evidence that the multimodal model achieves statistically superior classification performance compared to using only tabular data. Specifically, the tabular-only model achieved a mean AUROC of 0.79, with individual

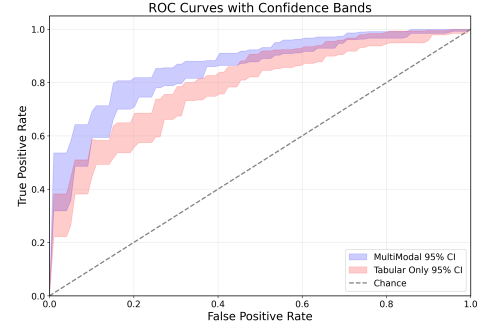


Fig. 3. **Mean ROC Curves with 95% Confidence Bands:** Mean Receiver Operating Characteristic (ROC) curves for the MultiModal model (blue) and the Tabular Only baseline model (red), plotted with their respective 95% confidence intervals (shaded regions), calculated across the 10 cross-validation folds. The dashed line indicates chance performance (AUROC = 0.5).

fold AUCs ranging from a minimum of 0.75 to a maximum of 0.84.

To further analyze the consistency of this improvement across the different data subsets used in cross-validation, the difference in AUROC was examined on a per-fold basis. Figure 4 directly visualizes the performance gain achieved by the multimodal approach over the tabular-only baseline across each fold. The bars represent the difference in AUROC score (Multimodal AUROC minus Tabular AUROC) for folds 0 through 9. Notably, the multimodal model achieved a higher AUROC score in every single fold, as indicated by the positive difference values for all bars. The mean improvement in AUROC across the 10 folds was 0.0653, demonstrating a consistent and significant advantage conferred by the integration of ultrasound image features alongside tabular data for this classification task across all evaluated data partitions.

VII. DISCUSSION

This study explored a multimodal deep learning approach for appendicitis diagnosis, motivated by the clinical practice of integrating diverse patient information and the potential for AI to synergize heterogeneous data sources, such as clinical records and ultrasound images. The multimodal model achieved an average AUROC of 0.86 on the evaluated dataset.

Previous work by Marcinkevičs et al., using Gradient Boosting Machines [35], reportedly reached a high AUROC of 0.85 on this same dataset using only tabular features (similar to the tabular input in this study, which also excluded ultrasound findings for this specific comparison). In contrast, the unimodal neural network in this work, processing only tabular

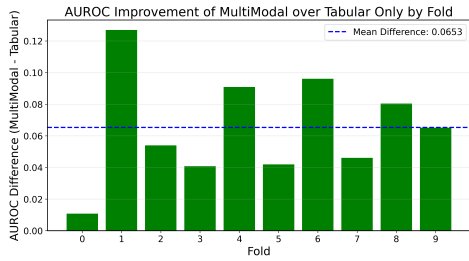


Fig. 4. **AUROC Improvement of MultiModal over Tabular Only by Fold:** Bar chart illustrating the difference in Area Under the Receiver Operating Characteristic Curve (AUROC) between the multimodal model and the tabular-only model for each of the 10 cross-validation folds. The vertical axis represents the absolute improvement (Multimodal AUROC - Tabular AUROC). The dashed blue line indicates the mean improvement across all folds (0.0653).

data, achieved an AUROC of just 0.79. This lower performance for the neural network on tabular data is somewhat expected, particularly considering the challenges deep learning models face with heterogeneous tabular data compared to traditional machine learning methods like Gradient Boosting Machines [18]. Comparing the previous work on Gradient Boosting Machines with the multimodal approach employed here reveals only a slight performance increase. This suggests that tabular data strongly influenced the final performance. Furthermore, this finding presents an interesting contrast to the work by Marcinkevičs et al., who achieved a notable AUROC of 0.80 using only ultrasound images processed through interpretable Concept Bottleneck Models [24]. In their approach, interpretable latent features—called concepts—are bottlenecked, meaning they can be accessed and analyzed at test time. This demonstrates that learning intermediate concepts is beneficial in medical imaging tasks. Their success underscores the potential of specialized architectures and possibly tailored preprocessing techniques for maximizing information extraction purely from images. While their approach offers advantages in interpretability, it also relies on clinicians' understanding of tuning underlying concepts for effective use in diagnosis. It represents a different strategy compared to the multimodal fusion approach employed in this study. Nonetheless, the present work demonstrates the feasibility of integrating raw image features, even if tabular data appears to dominate in this specific fusion setting. This supports the broader goal of developing automated CADx systems [12] that reduce dependence on manual feature extraction and may help decrease diagnostic uncertainty.

However, several limitations must be acknowledged. The model was developed using a single public dataset, necessitating external validation on diverse, multi-center data to assess generalizability [36]. I primarily explored concatenation and weighted sum fusion; other strategies like attention based [14] might yield different modality contributions or overall performance. Lastly, the inherent "black box" nature of the MLP and Residual Network (ResNet) components limits interpretability compared to models like Concept Bottleneck Models, posing a barrier to clinical trust.

Future research should prioritize external validation and

prospective clinical evaluation. Methodologically, exploring advanced fusion techniques and alternative architectures is a potential direction. Benchmarking against existing standards and unimodal baselines remains crucial. Incorporating interpretability methods is vital to address the black-box issue and facilitate clinical adoption. Furthermore, careful consideration of human-computer interaction is essential for successful clinical translation.

In conclusion, this study demonstrates the potential of end-to-end multimodal deep learning for appendicitis diagnosis using raw clinical and ultrasound data. While requiring further validation and refinement, the results contribute towards developing more automated, objective AI tools for managing this challenging clinical condition.

ACKNOWLEDGMENT

The authors would like to thank Dr. David Root and Dr. Jean-Pierre Teyssier at Agilent Technologies for the loan of the time-domain nonlinear measurement equipment and TriQuint Semiconductor for the donation of the transistors.

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Michael Roberg (S'09) received the B.S.E.E degree from Bucknell University, Lewisburg, PA, in 2003, the M.S.E.E. degree from the University of Pennsylvania, Philadelphia, in 2006, and the Ph.D. degree from the University of Colorado at Boulder in 2012. From 2003 to 2009, he was an Engineer with Lockheed Martin-MS2, Moorestown, NJ, where he was involved with advanced phased-array radar systems. His current research interests include high efficiency microwave PA theory and design, microwave power rectifiers, MMIC design, and high-efficiency radar and communication system transmitters. He is currently employed by TriQuint Semiconductor - Defense Products and Foundry Services in Richardson, TX working on wideband high efficiency GaN MMIC PA design.



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Tibault Reveyrand (M'07) received the Ph.D. degree from the University of Limoges, France, in 2002. From 2002 to 2004, he was a Post-Doctoral Scientist with CNES (French Space Agency). In 2005, he became a CNRS engineer at XLIM. His research interests include the characterization and modeling of RF and microwave nonlinear components and devices. Dr. Reveyrand was the recipient of the 2002 European GaAs Best Paper Award and is a member of the IEEE MTT-11 "Microwave Measurements" Technical Committee.



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Ignacio Ramos (S'12) received the B.S. degree in electrical engineering from the University of Illinois at Chicago in 2009, and is currently working toward the Ph.D. degree at the University of Colorado at Boulder. From 2009 to 2011, he was with the Power and Electronic Systems Department at Raytheon IDS, Sudbury, MA. His research interests include high-efficiency microwave power amplifiers, microwave DC/DC converters, radar systems, and wireless power transmission.



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Erez Avigdor Falkenstein (S'07), Haifa, Israel in 1979. He earned a "Handesaie" degree (associate degree) in electronics from Amal Handesaim School Hadera, Israel in 1999. From 1999 to 2003 he served in the Israel Defense Force as part of a technological unit. He has been at the University of Colorado at Boulder 2004 – 2012. He received concurrent MS/BS degrees in Electrical engineering 2010 and a Ph.D 2012 from the University of Colorado at Boulder. Since 2007 he has been involved with research as part of the active antenna group. Research emphasis: far field wireless powering for low power densities. Interests include Antenna design and characterization, modeling and measurement of nonlinear devices at microwave frequencies and power management. He is currently employed at Qualcomm, Incorporated, Boulder, CO.



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Zoya Popović (S'86-M'90-SM'99-F'02) received the Dipl.Ing. degree from the University of Belgrade, Belgrade, Serbia, Yugoslavia, in 1985, and the Ph.D. degree from the California Institute of Technology, Pasadena, in 1990. Since 1990, she has been with the University of Colorado at Boulder, where she is currently a Distinguished Professor and holds the Hudson Moore Jr. Chair with the Department of Electrical, Computer and Energy Engineering. In 2001, she was a Visiting Professor with the Technical University of Munich, Munich, Germany.

Since 1991, she has graduated 44 Ph.D. students. Her research interests include high-efficiency, low-noise, and broadband microwave and millimeter-wave circuits, quasi-optical millimeter-wave techniques, active antenna arrays, and wireless powering for batteryless sensors. Prof. Popović was the recipient of the 1993 and 2006 Microwave Prizes presented by the IEEE Microwave Theory and Techniques Society (IEEE MTT-S) for the best journal papers and the 1996 URSI Issac Koga Gold Medal. In 1997, Eta Kappa Nu students chose her as a Professor of the Year. She was the recipient of a 2000 Humboldt Research Award for Senior U.S. Scientists of the German Alexander von Humboldt Stiftung. She was elected a Foreign Member of the Serbian Academy of Sciences and Arts in 2006. She was also the recipient of the 2001 Hewlett-Packard (HP)/American Society for Engineering Education (ASEE) Terman Medal for combined teaching and research excellence.